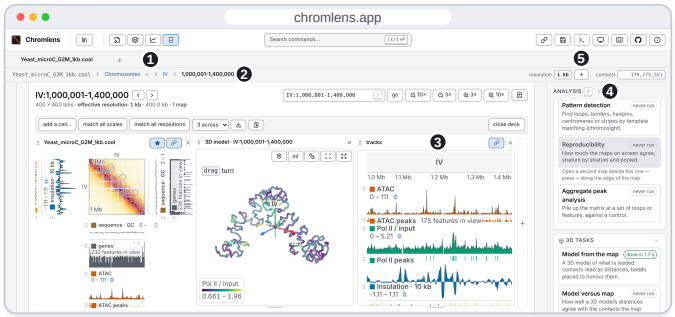
a A Chromlens session in a browser tab



❶ Files read in place, nothing uploaded
❷ Only the region and resolution in view
❸ A deck: 2D map, 3D model, 1D tracks
❹ Over 30 analyses, in a background worker
❺ Each step logged as its command line

b 1D, 2D and 3D data on one coordinate system

1D — Sequence, reads, signal, intervals

reads: FASTA, 2bit, BAM, VCF, bigWig, bedGraph, wig, BED, bigBed, GTF/GFF, genePred, bedMethyl, barChart, splice junctions, cytoBand

computes: peaks (yapc, IDR) · metaplots · track correlation · FRiP · signal by group · interval overlap · motif search · base composition · ori/ter

2D — Contact maps, interactions

reads: .cool, .mcool, .scool, .hic, HiC-Pro, BEDPE, interact, longrange, .pairs (binned on the fly)

computes: P(s) · insulation · directionality · Hi-C score · scalogram · compartments, saddle · loops, borders (Chromosight) · APA, pile-ups · virtual 4C · HiCRep · subsampling · Boost-HiC · inter-arm alignment · single-cell QC

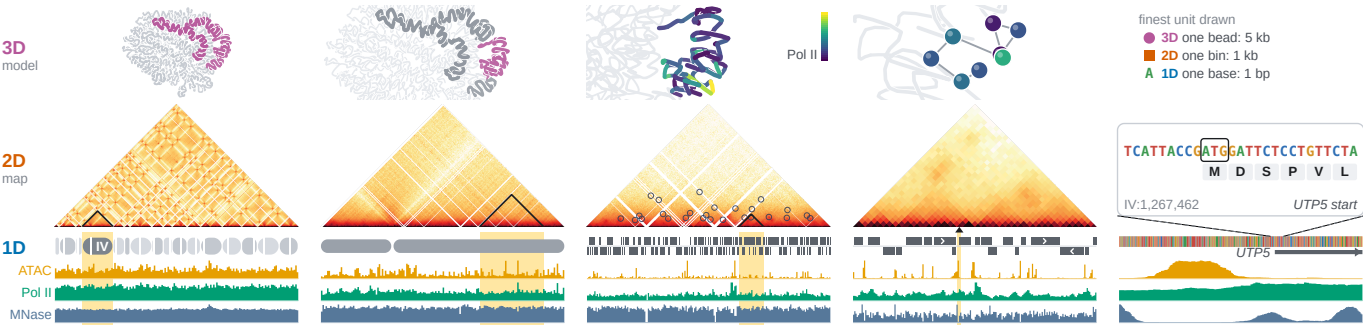
3D — Models

reads: g3d, 3dg, bed3d

computes: pastis-NB · ShRec3D · model vs map · spatial tracks

c One session, from the whole genome to single bases

Genome	×8	Chromosome IV	×4	Locus	×10	Gene	×80	Base pairs
12.1 Mb · 32-kb bins		1.53 Mb · 4-kb bins		400 kb · 1-kb bins · loops		40 kb · 1-kb bins, 5-kb beads		500 bp · single bases



d See ⇌ measure: every output is the next input, across dimensions

